Ultrastructural calcium simulations in neurons

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Summary

In this project we intend to study the calcium dynamics in reconstructed neurons, where the endoplasmic reticulum (ER), a multifunctional intracellular organelle of a neuron that consists of a complex three-dimensional network of connected endomembrane tubules, stacks and cisternae, is involved in various ion exchange processes. Of interest here are the three-dimensional spatiotemporal Ca^{2+} and Inositol-3-Phosphate (IP_3) dynamics in the intracellular space of neurons. We model these biochemical dynamics by a system of diffusion-reaction equations with highly nonlinear, ordinary differential equation flux-boundary conditions. Using finite volume discretization together with geometric multigrid methods (GMG) and finite difference methods on 1d reconstructions to model synapse loss with bidirectional electrical and ion coupling, we intend to run systematic numerical simulations of the calcium model on (1) Spine-Dendrite geometries with multispines and (2) full cell geometries under synapse loss. While only 1d morphological neuron data is directly available, our project requires a detailed 3d representation of the neuronal computational domain for multispine experiments and only 1d reconstructions for modeling bidirectional coupling for synapse loss. For this we developed an automated pipeline to generate the necessary domains. Thus, we (3) intend to study and reconstruct cells from the `www.neuromorpho.org` database using available HPC resources. In total, this project should positively impact the computational mathematics research community, as well as parallel computing and neuroscience fields.

1 Research Objectives

Computational Neuroscience is a rapidly developing research field that demands more and more computing resources for solving complex problems. Where in the past, models of signal processing in brain cells were strongly reduced – due to the lack of computing power and high resolution experimental data – they are now being developed in more detail. Emerging areas of research encompass large scale network simulations, three-dimensional simulations of neuronal signals (Xylouris et al. (2007); Xylouris (2013); Breit et al. (2016); Breit & Queisser (2018); Breit et al. (2018)) and multiscale modeling and simulation in neuroscience (Stepniewski et al. (2019)).

In order to resolve biochemical and electrical signals of neurons, based on physical first principles, models defined by sets of highly nonlinear and coupled partial and ordinary differential equations need to be solved numerically and efficiently. Recently, biophysical models of electrical and biochemical signaling, in particular calcium dynamics, have been developed and implemented in the multiphysics simulation platform uG4 (Vogel et al. (2013)). uG4 has been shown to scale optimally on massively parallel architectures (Reiter et al. (2014)). In various research projects the interplay between three-dimensional cell and organelle architecture and calcium signaling could be demonstrated (Grein et al. (2014); Wittmann et al. (2009); Queisser et al. (2011); Knodel et al. (2014); Breit & Queisser (2018); Breit et al. (2018)). From these projects two currently active ones emerged that focus on two distinct cellular scales: (1) the ultrastructural dynamics at contact points between two neurons (synapses/synaptic spines and multispines) and (2) whole cell calcium dynamics including the calcium exchange mechanisms at the cellular plasma membrane and the endoplasmic reticulum. The importance of studying calcium dynamics are manifold. For instance, spine and cellular calcium dynamics are implicated in the long-term effects of repetitive transcranial magnetic stimulation (rTMS), a technique used in clinical settings to treat e.g. schizophrenia. At the whole cell level calcium-induced calcium release dynamics controlled by the endoplasmic reticulum have been shown critical in learning, memory formation, and cell survival (West et al. (2002); Milner et al. (1998); Bading (2000); Hardingham et al. (1997)), as well as in neurodegenerative diseases, such as Alzheimer's and Parkinson's disease (Bezprozvanny (2009); Chan et al. (2009); Cheung et al. (2008); Green & LaFerla (2008)).

The objectives in this research proposal are to enable high-throughput numerical simulations to study the effect of a multitude of geometric and biological parameters on calcium signaling pathways in neurons. An active NIH funded research project on multiscale modeling of rTMS, in collaboration with Dr. Opitz (Engineering, University of Minnesota), Dr. Jedlicka (Computational Neuroscience, University of Giessen, Germany), and Dr. Vlachos (Neurophysiology, University of Freiburg, Germany), as well as Ph.D. research focusses on the effect of synapse loss on synapse to cell nucleus communication will be supported by the computing resources applied for below. Resources will support the following two research objectives:

1. At the level of synapses, synaptic spine morphology has been shown to change over time (Segal & Korkotian (2014); Maggio & Vlachos (2014); Verkhratsky (2004); Finch et al. (2012); Jedlicka & Deller (2017)). Along with this change, the intracellular cell organization adapts to ensure spine to dendrite coupling and calcium homeostasis in the spine head (Breit et al. (2018)). In the proposed research, spine geometries that have been reconstructed in the rTMS project mentioned above, will be used to run 3D ultrastructural calcium simulations while scanning a parameter space consisting of receptor and calcium pump densities in the plasma- and endoplasmic membrane. These systematic simulation runs will identify critical parameter sets at which switching of neuronal behavior can be observed; in particular, we will determine under what physiological and geometric conditions we have stable calcium propagation from the spine-to-dendrite and wave propagation through the length of the dendrite. Input for these simulations is voltage data derived from rTMS-induced electrical cell activity, which then drive voltage dependent calcium channels which are relevant for synaptic plasticity, stable calcium wave propagation, cell adaption, and survival.

2. Whole cell calcium dynamics will be run by employing a 3D cell reconstruction method (Mörschel et al. (2017); Grein (2020)) which uses 1D neuroanatomical data that is readily available from morphology databases such as NeuroMorpho.org (Ascoli

et al. (2007)). This technique will allow high-throughput reconstruction of a large number of neurons on which parallel computations of cellular calcium signaling can be performed. Again, scanning a parameter space consisting of geometric parameters (e.g. ER size) and biophysical parameters, such as receptor densities, will ideally produce a rigorous understanding of critical states at which neurons loose their ability to reliably transmit calcium signals to the cell nucleus (which then triggers pathological states).

2 Computational Methology

The simulations of spatio-temporal dynamics outlined in Sec. 4 will be carried out with the multiphysics simulation package uG4 (Vogel et al., 2013; Heppner et al., 2013). This decades-old project focusses on solving partial differential equations on unstructured grids and has been successfully used in applications from diffusion-advection systems, Navier-Stokes equations, and mechanics applications. In particular the code has been developed to run very well on highly parallel architectures (Vogel et al., 2013; Heppner et al., 2013). To document uG4's general scaling behavior, we performed benchmark scaling simulations to solve the second order elliptic partial differential equation $\Delta\Phi = 0$, the Laplace equation with Dirichlet boundary conditions. The PDE problem was solved on the unit cube with seven levels of regular grid refinement which contains 16,974,593 DoFs. The scaling is shown in Table 1 of **Section General scaling properties of uG4** in the *performance* document. For this proposal we will be making use of first-order finite volume discretization methods for the underlying systems of partial differential equations. The resulting system of linear equations will be solved using a geometric multigrid method (GMG), combined with an implicit time-stepping scheme. For more details we refer to Sec. 4.

3 Application efficiency

The SDSC Expanse CPU resource is suitable for our research because it allows high capacity computing and a quick turnaround on small to medium-scaled jobs of shorter runtimes, which makes it ideal for running the parameter sweep simulations in Sec. 4. The large shared memory per node (256 GiB) allows us to load large computational grids which would not be possible with the smaller shared memory per node on comparable resources. Comparing our demands with the XSEDE Resource Selector[1] confirmed our pre-selected resources. Application efficiency is documented in the attached file *performance*.

4 Computational Research Plan

4.1 Synaptic spine dynamics

The model: Synaptic spines are protrusions from dendrites and form contacts with chemically connected cells. Examples of 3D-reconstructed spines are shown in Fig. 1. The spine-interior often contains an organelle, the spine apparatus (SA), which is a continuous extension of the endoplasmic reticulum (ER) in the dendrite. The ER is a multifunctional

[1] Information was retrieved from `https://portal.xsede.org/allocations/resource-info`

intracellular organelle of a neuron, which consists of a complex three-dimensional network of connected endomembrane tubules, stacks and cisternae, is involved in various ion exchange processes, Breit & Queisser (2018), and ultimately responsible for synaptic plasticity. The SA and ER membrane harbors calcium exchangers that operate between the cytosol and ER domains. Typically, these are IP3-receptors, Ryanodine-receptors, and SERCA pumps. On the plasma membrane additional calcium exchangers operate between the extra- and intracellular space, namely Plasma Membrane Calcium Pumps (PMCA), sodium-calcium-exchangers (NCX) and voltage dependent calcium channels (VDCCs). The relevant quantities that need to be tracked on the 4D space/time cylinder are intracellular calcium Ca^{2+}, the molecule IP_3, which activates IP3-receptors together with Ca^{2+}, and mobile buffers, such as calbindin-D28k. One can define the following diffusion-reaction system first with the interaction between cytosolic Ca^{2+} calbindin-D_{28k} (CalB) is described by

$$\mathrm{Ca}^{2+} + \mathrm{CalB} \underset{\kappa_b^-}{\overset{\kappa_b^+}{\rightleftharpoons}} [\mathrm{CalBCa}^{2+}]. \tag{1}$$

The full domain equations for cytosolic calcium and calbindin-D_{28k} are thus given by

$$\frac{\partial c_c}{\partial t} = \nabla \cdot (D_c \nabla c_c) + \left(\kappa_b^- \left(b^{\text{tot}} - b\right) - \kappa_b^+ \, b \, c_c\right), \tag{2}$$

$$\frac{\partial b}{\partial t} = \nabla \cdot (D_b \nabla b) + \left(\kappa_b^- \left(b^{\text{tot}} - b\right) - \kappa_b^+ \, b \, c_c\right) \tag{3}$$

Exponential IP_3 decay towards a basal IP_3 concentration p^r in the cytosolic space is modeled by a reaction term that is added to the IP_3 diffusion equation, leading to the diffusion-reaction equation

$$\frac{\partial p}{\partial t} = D_p \Delta p - \kappa_p (p - p^r) \tag{4}$$

for IP_3 in the cytosolic domain. Endoplasmic Ca^{2+} dynamics are modeled by simple diffusion

$$\frac{\partial c_e}{\partial t} = D_c \Delta c_e \tag{5}$$

in the endoplasmic domain. Nonlinear Neumann boundary conditions, which are defined by the calcium channel and pump models. For the specifics of the ordinary differential equations that define PMCA, NCX, RyR, and SERCA pumps we refer to Breit et al. (2018). The VDCCs are plasma membrane channels which release calcium into the cytosol once a voltage is applied to the plasma membrane. The flux equations for VDCCs are given by

$$j_{vdcc} = G(V, t) F(V, \Delta[\mathrm{Ca}^{2+}]) \tag{6}$$

where G is the gating function and F is the flux function (Borg-Graham (1999)).

Expected outcome: A summary of our progress is given the *progress report* document and our results are conveyed in a manuscript which is in preparation. The results explained in the summary verified that spine morphologies fall into distinct clusters based on the critical RyR density for reliable calcium propagation to the dendrite. This critical RyR density depends on the shape and size of the morphology (see Fig. 1); in particular, we found from our simulations that changes in spine neck diameter affects the critical

RyR density. Meaning that spine-neck diameter is a determinant of *all-or-nothing* spine-to-dendrite calcium propagation. We will continue simulating calcium dynamics on a range of spine morphologies (which vary in shape and size, see Fig. 1) and varying the parameters of the model equations (see Breit et al. (2018) for specific parameter sets). Thus, we will be able to analyze the influence of cell and organelle morphologies on the calcium coupling behavior between spine and dendrite, which will provide novel insight into long-term potentiation of synapses and the role VDCCs play in spine-to-dendrite calcium coupling. We hypothesize there exists a particular pattern of synaptic activation and VDCC activation through backpropagating action potentials induced by rTMS that will ensure stable calcium wave propagation along the dendrite. We also hypothesize that the interplay between VDCC and synapatic activation is affected by the geometry of the ER, i.e. proximity of the ER membrane to the PM and cytosolic Ca^{2+} may induce further calcium release into the cytosol.

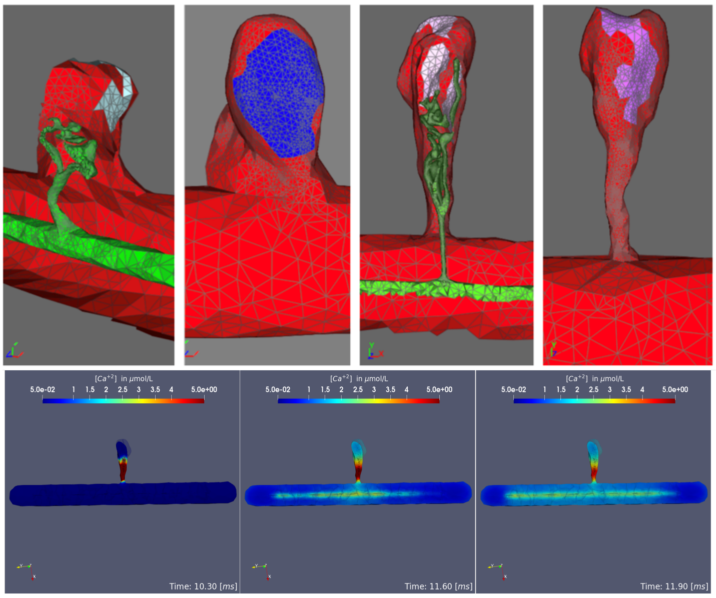

Figure 1: **Spine reconstructions and simulations:** (Top) Examples of spine reconstructions in collaboration with Dr. Vlachos (University of Freiburg, Germany). Reconstructions show variability in spine shape (red) as well as ER organization (green). (Bottom) Time sequence of a calcium simulation on reconstructed spine and dendrite geometry under VDCC activation using XSEDE resources. Color scheme: red: max. calcium concentration, blue: min. calcium concentration

Numerical method: To computationally solve the model problem, the triangular surfaces meshes of synaptic spines, which have been generated from electron microscopy data (in collaboration with Dr. Vlachos, University of Freiburg, Germany) will be used to produce tetrahedral volume meshes using ProMesh (Reiter (2012)). Using a first order finite volume method Eymard et al. (2000) the model equations are discretized on the previously generated surface and volume meshes. The nonlinear parts of the model, i.e. the Neumann boundary conditions are linearized with a Newton iteration. The resulting system of linear equations are solved by a geometric multigrid (GMG) method which is preconditioned using Bi-CGSTAB. The GMG solve of the linear system uses 3 iterations of pre- and post- smoothing, both of which use the *incomplete LU* factorization of the system matrix. We also set the *SuperLU* library as the base solver for our large, sparse, nonsymmetric matrix problem. A backward Euler method is used to solve the problem in time. The proposed numerical methods are available in uG4, which will be used for all proposed computations.

Estimate of proposed simulation runs: Next we describe the simulations we will perform for the above project and anticipated resources. The accuracy of the simulations relies on the following: employing physiologically correct geometries, running the simulations on geometries processed through several levels of grid refinements, running the simulations with small time step size Δt to capture the calcium dynamics in the cytosol and ER volume, and running the simulations with long end times. The studies we will perform are as follows:

1. `Single pulse runs` – We will continue to run simulations with active ER having ryanodine receptors (RyR) and SERCA pumps activated. For these simulations we run a parameter sweep on the density of RyR on the ER, this will establish the critical RyR density for calcium to be released from the ER into the cytosol. These simulations will be an expansion of our previous work (from the prior proposal) in that we will also take into account release of calcium by VDCCs. Similar to our prior work we sweep the RyR density value from 1 to 6 RyR per μm^2 at increments of 0.05, this amounts to 100 simulation runs, per cell. These simulations have an end time of 20 milliseconds since coupling of calcium in the dendritic region of the cell has been observed during the first 20 milliseconds of spine activation. This simulation set has 40,000 time steps with $\Delta t = 0.5$ μs.

2. `Low threshold calcium pulse runs` – We will run studies for at least 1 second to observe the coupling effects between the spine region of the neuron and the dendrite region of the neuron for a fixed RyR density and different calcium influx pulse frequencies. These particular simulations will be performed with supercritical RyR density, each at 1 Hz, 3 Hz, 5 Hz, 10 Hz, and 20 Hz frequencies. This totals to 5 simulations, each with end time of 1 second, $\Delta t = 0.5$ μs, this equates to $\frac{1.00}{0.5 \times 10^{-6}} = 2{,}000{,}000$ time steps. We will also be setting VDCC activation by driving the VDCCs at minimum voltage gate activation, i.e. using the minimum voltage V_m to induce calcium release by the VDCC.

3. `TMS-induced backpropagating action potential runs` – We will also run studies for at least 0.25 seconds to observe the coupling effects between the spine region of the neuron and dendrite; with fixed RyR density (at critical) and fixed pulse frequency. In this study we will vary the voltage input at voltage dependent calcium channels (VDCCs) to determine when a stable calcium wave is propagated along the dendrite. The voltage will be varied from -70 mV to 70 mV at increments of 10 mV (fifteen voltage levels) for each spine-dendrite geometry. This is $9 \times 15 = 135$ simulations, each simulation with time step $\Delta t = 0.5$ μs, this equates to $\frac{0.25}{0.5 \times 10^{-6}} = 500{,}000$ time steps per simulation. We will also use the data from these simulations to determine an appropriate minimum voltage to induce calcium release by VDCCs.

We have 9 different spine-dendrite geometries that will undergo each of the above simulations. Currently, with the XSEDE trial account, we ran these simulations with 0 refinement (20,605 Degrees of Freedom – DoFs) and 1 refinement (164,840 DoFs) on the Spine-Dendrite geometry and time step size $\Delta t = 0.5$ μs. In Fig. 3 of the *performance* document we show strong scaling for a geometry *without ER*. We ran a simulation with 164,840 DoFs across

512 CPU cores with compute time of 4.69 minutes = 0.0781 hours and with simulation endtime 0.02 seconds. Therefore the number of core hours = 512 $cores \times 0.0781$ $hours = 40$ core-hours. In order to capture accurate results we will process geometries with at least 1 level of refinement, with and without ER geometry present. In Table 1 we indicate the anticipated compute time requirements for the sub-projects. The total number of runs is determined by (#runs/cell) $\times$ (#cells); for instance, the `TMS runs` are 15 simulations and there are 9 cell geometries, therefore, we have (15 runs/cell) $\times$ (9 cells) = 135 runs. Then the number of core-hours are computed by (#runs) $\times$ (core-hours/run); therefore, for `TMS runs` we have (135 runs) $\times$ (40 core-hours/run) = $5,400$ core-hours for a 0.02 s endtime, this becomes 67,500 core-hours for 135 runs with 0.25 s endtime. Similarly for the `Single Pulse runs` we have $(6-1)/0.05 = 100$ runs per cell therefore we have $100 \times 40 = 4000$ core-hours per cell. For the `Low-Threshold Pulse runs` we need to take into account the number of time steps since the end time is 1 second, we have $\frac{40 \text{ core-hours}}{40000 \text{ time-steps}} \times 2000000$ time-steps = $2,000$ core-hours per run. Concerning storage requirements: we have two types of data output, text files in .txt format that contain the average calcium concentration in the volume and surface regions of the cell geometry and Visualization Toolkit files (vtk) for rendering and visualizing data in 3 dimensions. We will be generating .vtk output for the `Low-Threshold Pulse runs` with active RyR & SERCA pumps and we anticipate that each vtk file is 0.24 GiB at 1 refinement level. The `Low-Threshold Pulse` simulations run for 1 second and total 2 million time steps but we will require only 100 vtk time samples (equally distributed in time) out of the 2 million time steps. For the `Low-Threshold Pulse runs` with active RyR & SERCA pumps we will require $\left(0.24\frac{\text{GiB}}{\text{file}}\right) \times$ (100 files) $\times$ (9 cells) $\times$ (5 freq.) = **1,080 GiB** of vtk storage for all nine cells and five frequencies. For the `TMS runs` each simulation run will produce 27,500 kB (for end time of 0.25 seconds) of data in the form of text files, at 15 runs for 9 cells this equates to $\left(27,500\frac{\text{kB}}{\text{run}}\right) \times$ (15 runs) $\times$ (9 cells) = 3.7 GiB. For the RyR & SERCA pump (`Single Pulse`) simulations, we will be performing 900 simulation runs: $\left(2750\frac{\text{kB}}{\text{run}}\right) \times$ (900 runs) $\times$ (9 cells) = 22.2 GiB. For the `Low-Threshold Pulse` simulations we have a run time of 1 second which equates to 137,500 kB per run; therefore, $137,500 \times 15 \times 9 = 18.6$ GiB. Therefore, in total we anticipate $3.7+22.2+18.6 =$ **44.5 GiB** for text data; hence, for all spine simulations we anticipate $44.5 + 1,080 = 1.1$ TiB. If we include ER geometry and dynamics this will incur more storage; therefore, we anticipate per month approximately **1.5 TiB** after which all data will be copied to local storage.

4.2 Whole cell dynamics

The model: While subproject 4.1 focusses on the ultrastructure of individual synaptic spines, this subproject is defined on the whole cell level. The aim of this project is to study the importance of spatio-temporal synaptic activity in relaying stable calcium signals to the cell nucleus. The calcium model equations are the same as in Sec. 4.1. The computational meshes are derived by a method developed previously in Grein (2020). The ER will be modeled as a neuron-within-neuron and can be generated alongside the surface reconstruction of a neuron. The volumetric mesh is then generated with ProMesh (Reiter (2012)). The reconstruction pipeline is illustrated in Fig. 2. Simulations will be run by initializing the calcium model with synapses. Synapses will be placed at distinct locations on the dendritic tree. Using standard synapse models calcium influx will be modeled as time-dependent Neumann boundary conditions.

Expected outcome: Calcium signals relayed to the cell nucleus by intracellular calcium waves are crucial for learning, synaptic plasticity as well as cell survival, cf. (West et al. (2002); Milner et al. (1998); Bading (2000); Hardingham et al. (1997)). Neuropathologies, like Alzheimer's or ALS, impact calcium dynamics and disrupt calcium pathways (Bezprozvanny (2009); Chan et al. (2009); Cheung et al. (2008); Green & LaFerla (2008)) by loss of synapse synchronicity and/or synaptic strength.
In the previous project period we studied the necessary conditions for neurons to reliably evoke an intracellular calcium wave to transmit a signal to the cell nucleus. In particular we focussed on the effect of synaptic strength, distribution of synapses, and number of total synapses in synapse loss experiments. We hypothesized that the spatio-temporal synaptic input patterns are relevant for synapse-to-nucleus communication. To substantiate our preliminary findings, additional parameter sweeps will become necessary. In particular we intend to run simulations with a more diverse set of spatial synapse profiles and different temporal profiles, to test the influence of synapse clustering and synchronous vs. asynchronous activation on the possibility to evoke a stable or abortive intracellular calcium wave. Additionally, morphologically distinct neurons will need to be analyzed, since we recognized different synaptic stimulation patterns are required to evoke an intracellular calcium wave in different cell types. We further hypothesize that the morphological and biophysical parameters are important for stable communication. Thus, we plan to vary relevant receptor densities of the RyR and IP_3 receptors, activation kinetics of synapses to determine threshold parameter sets that distinguish between stable and unstable synapse-to-nucleus communication. The results of this study could contribute to new insight into pharmacological intervention during synapse loss, which is a major contributor in neurodegenerative diseases.

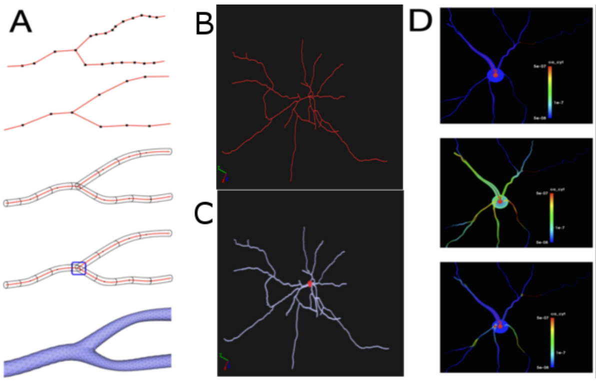

Figure 2: **3D reconstruction pipeline**: (A) Illustration of cell surface reconstruction using graph and diameter information available from NeuroMorpho.org. The reconstruction pipeline begins with cellular graph information (B) and constructs a watertight surface triangulation (C), using the method presented in (Grein (2020)). (D) Time evolution of a whole cell calcium simulation.

Numerical method: The numerical method for this sub-project is identical to Sec. 4.1.
Estimate of proposed simulation runs: To study the cellular calcium dynamics from synapses to the cell nucleus, three computational setups will be used. Representative computational meshes, which will be used in the studies outlined below and used in the strong and weak scaling benchmarks have roughly 451,485 DoFs (see Tab. 3, 4) with run times of roughly 5 minutes for 1,000 time step on 3,074 processors.

1. `Variation of location of synaptic stimulation regions` – We will use 10 representative cells from morphologically distinct regions from the NeuroMorpho.org database and generate the 3D cell and ER geometries using the approach outlined

above. The number of synapses determined in the previous study which reliably triggers a calcium wave will be placed on the dendrites of neurons and we will determine whether or not stable calcium waves can be triggered by varying the dendrite diameters. For the dendrite diameter a small, medium and large endoplasmic reticulum will be considered. For these simulation setups a simulation time of 1 second is necessary to track calcium waves through the entire neuron. With a timestep of 0.1 ms and 3 possible endoplasmic reticulum sizes this will amount to $3 \times 10 \times \frac{1{,}000}{0.1} = 300{,}000$ time steps for one specific synaptic stimulation pattern. We will need to address whether distribution patterns can impact wave generation in different brain regions. To address the question how spatial variation of synaptic stimulation will affect calcium waves, 5 distribution patterns will be tested: homogeneously distributed synapses 1) close to soma, 2) on distal dendrites, 3) on proximal dendrites, and clustered synapses (non-homogeneous) 4) on distal dendrites, and 5) proximal dendrites. In total $3 \times 10 \times \frac{1{,}000}{0.1} = 300{,}000$ time steps will be necessary. Using the benchmark data above, this yields $\frac{300{,}000}{1{,}000} \times \frac{5}{60} \times 3{,}072 = 76{,}800$ core hours.

2. `Variation of synaptic activation strength` – Using the same 10 neurons we will vary receptor densities for IP_3 receptors and RyRs to determine the conditions under which stable synapse-to-nucleus calcium communication is observed. For this study a simulation time of 2 seconds is necessary. We will test 3 settings: Low, intermediate and high receptor densities. Thus, in total we expect $3 \times 10 \times \frac{2{,}000}{0.1} = 600{,}000$ time steps, which yields $\frac{600{,}000}{1{,}000} \times \frac{5}{60} \times 3{,}072 = 153{,}600$ core hours.

3. `Synapse synchronicity` – We will employ 10 representative cells to study the effect of synapse synchronicity on synapse-to-nucleus calcium communication. For this we subdivide the parameter space into a no synchronicity, 50% synchronicity and full synaptic synchronicity. Synaptic synchronicity is known to be important for generating enough calcium influx to trigger calcium waves, important for communication in neurons from the distal to the proximal regions. From prior studies we know that abortive waves can be reliably detected within 1 s of simulation, thus when setting our time step to 0.1 ms to resolve action potentials, $10{,}000$ time steps are required for one neuron. In total, for 10 neurons with 3 states and 10,000 time steps for each simulation a total of $3 \times 10 \times \frac{1{,}000}{0.1} = 300{,}000$ time steps is required. Using the above benchmark data yields $\frac{300{,}000}{1{,}000} \times \frac{5}{60} \times 3{,}072 = 76{,}800$ core hours.

For a summary of representative simulation runs, see Table 1 with anticipated compute time (core-h / SUs) requirements for this sub-project. Storage requirement for one VTK file is roughly 1 MiB per time step. This yields, for all three studies, a total of 1.2 TiB.

4.3 Summary

The resources required for the two projects outlined in Sec. 4.1 and 4.2 are summarized in Tab. 1. The graduate student James Rosado will be conducting all runs related to Sec. 4.1 and Stephan Grein will be responsible for all runs for Sec. 4.2. The postdoc Qingguang Guan and PI Queisser will support and advise efforts in both projects.

All participants in the project will have access to a shared file in which an estimated demand for compute time for the following project month can be submitted. James Rosado will coordinate the internal distribution of compute time for the subsequent month by weighing the individual demand against milestone deadlines, demand for conference

Compute time requirements for **SDSC Expanse**						
Sub-project	Type of run	# runs	# Simulation Time /run [sec.]	Wall time /run [h]	# core-h/ run	Total [core-h] (SU)
Synaptic spine dynamics	TMS-induced	135	0.25	1.00	540	67,500
Synaptic spine dynamics	Single Pulse	900	0.02	1.00	40	36,000
Synaptic spine dynamics	Low-Threshold Pulse	45	1.00	1.00	2,000	90,000
Whole cell dynamics	Variation of regions	50	1.00	0.5	3,072	76,800
Whole cell dynamics	Receptor densities	30	1.00	0.5	3,072	153,600
Whole cell dynamics	Synchronicity	60	1.00	0.5	3,072	76,800
TOTAL						**500,700**

Table 1: Total number of anticipated SUs on SDSC Expanse for the two requested sub-projects as outlined in sections 4.1, 4.2.

contributions, theses finalizations and other constraints. Thus, contingent overflow as well as monthly over-demand can be minimized.

5 Justification of Resources Requested

The charge unit for SDSC Expanse is the Service Unit (SU). One SU corresponds to the use of one compute core for one hour (The minimum charge for a job longer than 10 seconds is 1 SU). Accordingly the requested SUs for the sub-projects in this proposal are summarized in Tab. 1, which have been determined by the following equation:

$$SUs = \sum Runtime(Walltime)\ [h] \quad \times \quad \#\ Cores \quad \times \quad \#\ Runs$$

The total storage requirement for the two sub-projects is **4.8 TiB**, in which the sub-project 4.1 requires **2 TiB** and sub-project 4.2 requires **2.8 TiB** of storage.

6 Additional considerations

Dr. Queisser (PI) and his research group has a long-standing experience in Scientific Computing. Prior to Dr. Queisser's relocation to the US from Germany, his group was awarded resources on the JUQUEEN and JUWELS supercomputers in Jülich, Germany over the period of 6 consecutive years. Currently, Dr. Queisser's postdoctoral researcher Dr. Guan and graduate student James Rosado are actively working on the subproject 4.1 which is funded through a recently awarded NIH grant. Both have experience in running code on remote parallel compute clusters. Furthermore, graduate student Stephan Grein is involved in the subproject 4.2. James Rosado and Stephan Grein ran the presented XSEDE benchmarks. The requested SU are relevant for carrying out the NIH project and for the completion of Grein's dissertational research. This project will support 70% of the National Health Institute (NIH) grant *CRCNS: Collaboration toward an experimentally validated multiscale model of rTMS*. The remaining 30% are requested to support the completion of two manuscripts in preparation and ongoing PhD research. Our group has access to the smaller-sized parallel computing system Owlsnest operated by Temple University, Philadelphia, with roughly 180 general computing CPUs plus some GPU capabilities on other nodes. While developing on local multicore machines, then porting to the local compute cluster Owlsnest for testing, debugging and running smaller batch parameter sweeps is feasible on the local infrastructure, we require access to larger machines for running production code simulations to validate publishable research results. These simulations are beyond what we can run locally under the given constraints.

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